



STRONG CONVERGENCE THEOREMS FOR GENERALIZED BREGMAN NONEXPANSIVE MAPPINGS IN BANACH SPACES

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ABSTRACT. The concept of ∇g -fixed point for nonlinear mappings in Banach spaces is introduced. We prove strong convergence theorems for generalized Bregman nonexpansive and Bregman generalized- ∇g -nonexpansive mappings. Different from the usual fixed point results, we also verify ∇g -fixed point results for generalized- ∇g -nonexpansive mappings in Banach spaces. Finally, we discuss the ∇g -generalized mixed equilibrium problem and prove new results beyond the usual ones in the literature.

1. INTRODUCTION

Along all lines of the article, we stand $\mathbb{N}$ and $\mathbb{R}$ as the set of natural numbers and the set of real numbers, respectively. Also, the set of extended real numbers will be shown by $\overline{\mathbb{R}}$, that is $\overline{\mathbb{R}} = [-\infty, +\infty]$. In the whole paper, we suppose a Banach space $(E, \|\cdot\|)$ has a dual space E^* and we indicate the duality pairing by $\langle u, u^* \rangle$, $\forall u \in E$ and $u^* \in E^*$. The strong convergence and the weak convergence of a sequence $\{x_n\}_{n \in \mathbb{N}} \subset E$ to $x \in E$ are indicated by $x_n \rightarrow x$ and $x_n \rightharpoonup x$, respectively. Let C be a nonempty, closed, and convex subset of E such that $\nabla g(C)$ is closed and convex, where $\nabla g : E \rightarrow E^*$ is the gradient mapping on E . Let ψ be a mapping from $\nabla g(C)$ to $\mathbb{R}$, h be a bi-function from $\nabla g(C) \times \nabla g(C)$ to $\mathbb{R}$, Φ be a nonlinear mapping from C to E^* . The ∇g -generalized mixed equilibrium problem is to find an element $v \in C$ such that

$$h(\nabla g(v), \nabla g(y)) + \psi(\nabla g(y)) - \psi(\nabla g(v)) + \langle \Phi v, y - v \rangle \geq 0 \quad \text{for all } y \in C.$$

The set of solutions of the ∇g -generalized mixed equilibrium problem is denoted by $\text{BGMEP}(h, \psi, \Phi)$. It is well known that the class of generalized mixed equilibrium problems contains, as special cases, numerous important classes of nonlinear problems such as variational inequality problems, optimization problems, equilibrium problems, and so on (see, e.g., [5, 12, 19, 20] and the references contained therein). Let $T : C \rightarrow E$ be a *nonexpansive mapping*. We denote the set of fixed points of T by $F(T)$.

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For several years, many authors have studied the problem of obtaining a common element in the set of solutions of equilibrium problems and the set of fixed points of nonexpansive mappings from E to E in the setting of real Banach spaces. In 2008, Peng and Yao [30] studied the problem of obtaining a common element in the set of solutions of a generalized mixed equilibrium problem and the set of fixed points of a nonexpansive mapping in a real Hilbert space H . To extend this result to more general classes of nonlinear mappings than nonexpansive ones, the concepts of generalized nonexpansive mappings and relatively nonexpansive mappings have been introduced and studied by several authors (see, e.g., Chidume et al. [9], Qin et al. [32], and the references therein).

Let E be a real normed space with dual space E^* . A mapping $\Phi : E \rightarrow E^*$ is called *monotone* if $\langle \Phi x - \Phi y, x - y \rangle \geq 0$, $\forall x, y \in E$. A mapping $\Phi : E \rightarrow E$ is called *accretive* if $\langle \Phi x - \Phi y, j(x - y) \rangle \geq 0$, $\forall x, y \in E$, for some $j(x - y) \in J(x - y)$. Numerous authors have extensively studied the class of accretive operators. To solve the equation $\Phi x = 0$, where $\Phi : E \rightarrow E$ is an *accretive operator*, Browder introduced the mapping $T : E \rightarrow E$ defined by $T := I - \Phi$, where I is the identity mapping on E , and called it *pseudocontractive*. It is clear that the solutions of $\Phi x = 0$ in this case correspond to fixed points of T . Consequently, approximating zeros of accretive operators can be done by approximating fixed points of pseudocontractive mappings. This fixed point technique, however, is not applicable when $\Phi : E \rightarrow E^*$ is a monotone map.

Let $Y \subset E$ be nonempty and $\mathcal{P}(Y)$ be the set of all nonempty subsets of Y . We set

- $\mathcal{C}(Y) := \{W \in \mathcal{P}(Y) : W \text{ is nonempty, closed}\},$
- $\mathcal{K}(Y) := \{W \in \mathcal{P}(Y) : W \text{ is nonempty, compact}\},$
- $\mathcal{C}v(Y) := \{W \in \mathcal{P}(Y) : W \text{ is nonempty, closed and convex}\},$
- $\mathcal{C}b(Y) := \{W \in \mathcal{P}(Y) : W \text{ is nonempty, closed and bounded}\},$
- $\mathcal{SBGNR}(Y) := \{W \in \mathcal{P}(Y) : W \text{ is sunny, Bregman, generalized, nonexpansive and retract}\}.$

NST-Condition (Klin-eam et al. [21]). Let $\{T_n\}_{n \in \mathbb{N}}$ and Θ be two families of generalized nonexpansive mappings from C into E such that $\bigcap_{n=1}^{\infty} F(T_n) = F(\Theta) \neq \emptyset$, where $F(T_n)$ is the set of fixed points of T_n and $F(\Theta)$ is the set of common fixed points of all mappings in Θ . A sequence $\{T_n\}_{n \in \mathbb{N}}$ from C to E is said to satisfy the NST-condition with Θ if for each bounded sequence $\{x_n\} \subset C$, $\lim_{n \rightarrow \infty} \|x_n - T_n x_n\| = 0$ implies $\lim_{n \rightarrow \infty} \|x_n - T x_n\| = 0$ for all $T \in \Theta$.

Many research articles concerning convex minimization problems are based on the algorithms in different style. Some of them do not contribute acceleration of convergence as one can discover in [21, 33] and some others. In order to solve minimization problems and achieve the acceleration of the process, Polyak [31] introduced an algorithm with an inertial item which enjoys a very high rate of convergence of the sequence produced by the algorithm. Many researchers [1, 9, 10, 12, 14, 18, 21, 23, 33, 34] have investigated the inertial-type algorithms and have found some weak and strong convergence results. In 2012, Klin-eam et al. [21] investigated the following CQ algorithm [24] for generalized nonexpansive mappings in a Banach

space:

$$(1.1) \quad \begin{cases} x_1 = x \in C, & C_0 = Q_0 = C, \\ u_n = \alpha_n x_n + (1 - \alpha_n) T_n x_n, \\ C_n = \{z \in C_{n-1} \cap Q_{n-1} : \phi(u_n, z) \leq \phi(x_n, z)\}, \\ Q_n = \{z \in C_{n-1} \cap Q_{n-1} : \langle x - x_n, Jx_n - Jz \rangle \leq 0\}, \\ x_{n+1} = R_{C_n \cap Q_n} x, \quad \forall n \geq 1, \end{cases}$$

where J is the normalized duality mapping on E , $\{T_n\}_{n \in \mathbb{N}}$ satisfies NST-condition and $\{\alpha_n\}_{n \in \mathbb{N}} \subset [0, 1]$ with $\liminf_{n \rightarrow \infty} \alpha_n(1 - \alpha_n) > 0$. They proved the following theorem:

Theorem 1.1 (Klin-eam et al. [21]). *Let E be a uniformly smooth and uniformly convex Banach space and C be a nonempty closed subset of E such that JC is closed and convex. Let $\{T_n\}_{n \in \mathbb{N}}$ be a countable family of generalized nonexpansive mappings from C into E and let Ω be a family of closed generalized nonexpansive mappings from C into E such that $\bigcap_{n=1}^{\infty} F(T_n) = F(\Omega) \neq \emptyset$. Suppose that $\{T_n\}_{n \in \mathbb{N}}$ satisfies the NST-condition with Ω . Then, the sequence $\{x_n\}_{n \in \mathbb{N}}$ generated by equation (1.1) converges strongly to $R_{F(\Omega)}x$, where $\phi : E \times E \rightarrow \mathbb{R}$ by $\phi(x, y) := \|x\|^2 - 2\langle x, Jy \rangle + \|y\|^2$; for all $x, y \in E$.*

Definition 1.2. A Banach space E enjoys the Kadec-Klee property, if for any sequence $\{u_n\}_{n \in \mathbb{N}} \subset E$ the following implication holds:

$$u_n \rightharpoonup u \in E \text{ and } \|u_n\| \rightarrow \|u\| \implies \|u_n - u\| \rightarrow 0.$$

Theorem 1.3 ([11]). *Let E be a uniformly smooth and strictly convex real Banach space with the Kadec-Klee property and dual space E^* . Let $A \in Cv(E)$ be fixed and $T_n : E \rightarrow E$, for $n \in \mathbb{N}$, be a countable family of generalized nonexpansive mappings, and Ω be a family of closed and generalized nonexpansive mappings from E to E such that $\bigcap_{n=1}^{\infty} F(T_n) = F(\Omega) \neq \emptyset$. Let $\{v_k\}_{k \in \mathbb{N}}$ be generated by:*

$$\begin{cases} v_0, v_1 \in E, & A_1 = E, \\ w_n = v_n + \gamma_n(v_n - v_{n-1}), \\ y_n = \alpha w_n + (1 - \alpha) T_n w_n, \\ A_{n+1} = \{v \in A_n : \phi(y_n, v) \leq \phi(w_n, v)\}, \\ v_{n+1} = R_{A_{n+1}} v_0, \quad \forall n \geq 1. \end{cases}$$

Assume that $\{T_n\}_{n \in \mathbb{N}}$ satisfies the NST-condition with Ω , $\{\gamma_n\}_{n \in \mathbb{N}} \subset (0, 1)$, and $\alpha \in (0, 1)$. Then, $\{v_n\}_{n \in \mathbb{N}}$ converges strongly to $R_{F(\Omega)}v_0$.

In a series of papers [9, 12, 14, 18, 23, 34], the authors investigated fixed point theorems for generalized nonexpansive mappings in uniformly smooth and uniformly convex Banach spaces. Most of these results are obtained under the Kadec-Klee property of the Banach spaces. In this paper, using Bregman distances, we employ a new technique beyond the Kadec-Klee property and provide new results in the fixed point theory of generalized Bregman ∇g -nonexpansive mappings in Banach spaces. Our results improvement some of corresponding ones in [11–14, 18, 21].

2. PRELIMINARIES

For a proper lower semi-continuous convex function $g : E \rightarrow (-\infty, +\infty]$, the *subdifferential* ∂g of g is defined by

$$\partial g(u) = \{u^* \in E^* : g(u) + \langle v - u, u^* \rangle \leq g(v), \forall v \in E\}$$

for all $u \in E$. For any proper lower semi-continuous convex function $g : E \rightarrow (-\infty, +\infty]$, the *conjugate function* g^* of g is defined by

$$g^*(u^*) = \sup_{u \in E} \{\langle u, u^* \rangle - g(u)\}$$

for all $u^* \in E^*$. We know that (cf. [36]) $g(u) + g^*(u^*) \geq \langle u, u^* \rangle$ for all $(u, u^*) \in E \times E^*$. $(u, u^*) \in \partial g$ is equivalent to

$$g(u) + g^*(u^*) = \langle u, u^* \rangle.$$

If $g : E \rightarrow (-\infty, +\infty]$ is a proper lower semi-continuous convex function, then $g^* : E^* \rightarrow (-\infty, +\infty]$ is a proper weak* lower semi-continuous convex function [36]. The norm-to-weak* continuity of ∇g is discussed in [4, Proposition 1.1.10] and the norm-to-norm continuity of ∇g is presented in [22, p. 508]. The set of all strictly convex, Gâteaux differentiable, proper and lower semi-continuous functions $g : E \rightarrow (-\infty, +\infty]$ is denoted by $\Lambda_1(E)$.

The convex function $g : E \rightarrow (-\infty, +\infty]$ is called *Fréchet differentiable* at $x \in E$ (see, for example, [4, 22]) if for all $\epsilon > 0$, there exists $\delta > 0$ such that $\|v - u\| \leq \delta$ implies that

$$|g(v) - g(u) - \langle v - u, \nabla g(u) \rangle| \leq \epsilon \|v - u\|.$$

A convex function $g : E \rightarrow (-\infty, +\infty]$ is said to be *Fréchet differentiable* if it is Fréchet differentiable everywhere. If a continuous convex function $g : E \rightarrow (-\infty, +\infty]$ is *Gâteaux differentiable*, then ∇g is norm-to-weak* continuous ([4]). Also, if g is Fréchet differentiable, then ∇g is norm-to-norm continuous ([22]). The mapping ∇g is said to be *weakly sequentially continuous* if $u_n \rightharpoonup u$ implies that $\nabla g(u_n) \rightharpoonup^* \nabla g(u)$ ([4, 22]). The function g is said to be *strongly coercive* if

$$\|u_n\| \rightarrow \infty \implies \frac{g(u_n)}{\|u_n\|} \rightarrow \infty.$$

It is known that if g is strongly coercive then the function g^* strongly coercive [36]. The function g is said to be bounded on bounded sets if $g(Y)$ is bounded for each bounded subset Y of E .

Let $g : E \rightarrow (-\infty, +\infty]$ be a convex and Gâteaux differentiable function. Then the Bregman distance [2, 7] of g is denoted by $D_g : E \times E \rightarrow \mathbb{R}$ and

$$D_g(u, v) = g(u) - g(v) - \langle u - v, \nabla g(v) \rangle, \forall u, v \in E.$$

In addition, we have

$$D_g(u, v) = -D_g(v, u) + \langle u - v, \nabla g(u) - \nabla g(v) \rangle, \forall u, v \in E.$$

We also define the function $D_{g^*} : E^* \times E^* \rightarrow \mathbb{R}$ by

$$D_{g^*}(u^*, v^*) = g^*(u^*) - g^*(v^*) - \langle \nabla g^*(v^*), u^* - v^* \rangle.$$

It is evident that $D_{g^*}(\nabla g(v), \nabla g(u)) = D_g(u, v)$, $\forall u, v \in E$. The Bregman distance [2, 7] enjoys the following three-point identity:

$$(2.1) \quad D_g(u, z) = D_g(u, v) + D_g(v, z) + \langle u - v, \nabla g(v) - \nabla g(z) \rangle, \quad \forall u, v, z \in E.$$

Let E be a Banach space and B_E be the unit ball of E . Let $B_r(0) := \{z \in E : \|z\| \leq r\}$ for all $r > 0$. Then a function $g : E \rightarrow (-\infty, +\infty]$ is said to be *uniformly convex* on bounded sets ([36]) if $\rho_r(t) > 0$ for all $r, t > 0$, where $\rho_r : [0, +\infty) \rightarrow [0, \infty]$ is defined by

$$\rho_r(t) = \inf_{u, v \in B_r(0), \|u-v\|=t, \beta \in (0,1)} \frac{\beta g(u) + (1-\beta)g(v) - g(\beta u + (1-\beta)v)}{\beta(1-\beta)}$$

for all $t \geq 0$. The function ρ_r is called the gauge of uniform convexity of g . The function g is also said to be *uniformly smooth* on bounded sets ([36]) if $\lim_{t \downarrow 0} \frac{\sigma_r(t)}{t} = 0$ for all $r > 0$, where $\sigma_r : [0, +\infty) \rightarrow [0, \infty]$ is defined by

$$\sigma_r(t) = \sup_{u \in B_r(0), v \in S_E, \beta \in (0,1)} \frac{\beta g(u + (1-\beta)tv) + (1-\beta)g(u - \beta tv) - g(u)}{\beta(1-\beta)}$$

for all $t \geq 0$.

The function g is said to be uniformly convex if the function $\delta_g : [0, +\infty) \rightarrow [0, +\infty]$, defined by

$$\delta_g(t) := \sup \left\{ \frac{1}{2}g(u) + \frac{1}{2}g(v) - g\left(\frac{u+v}{2}\right) : \|v-u\| = t \right\},$$

satisfies that $\lim_{t \downarrow 0} \frac{\delta_r(t)}{t} = 0$.

Remark 2.1. In particular, if $\Omega = \{T\}$, i.e., Ω consists of one mapping T , then $\{T_n\}_{n \in \mathbb{N}}$ is said to satisfy the NST-condition with T . It is obvious that $\{T_n\}_{n \in \mathbb{N}}$ with $T_n = T$, for all $n \in \mathbb{N}$, satisfies the NST-condition with $\Omega = \{T\}$.

Definition 2.2. For any $C \in \mathcal{Cv}(E)$, we recall the following different kinds of mappings discussed in [29].

- (1) A mapping $T : C \rightarrow E$ is called Bregman generalized nonexpansive if $F(T)$ is nonempty and $D_g(Tu, q) \leq D_g(u, q)$, for all $u \in C$, $q \in F(T)$.
- (2) A mapping $R : E \rightarrow C$ is called a retraction if $R^2 = R$. Furthermore, R is called sunny if $R(Ru + t(u - Ru)) = Ru$, for all $u \in E$ and $t \leq 0$.
- (3) An element C of $\mathcal{C}(E)$ is called a sunny Bregman generalized nonexpansive retract of E if there exists a sunny Bregman generalized nonexpansive retraction R from E onto C . This mapping will be denoted by R_C^g .
- (4) A mapping $T : C \rightarrow E$ is called closed if for any $\{u_n\}_{n \in \mathbb{N}} \subset C$ such that $u_n \rightarrow u^*$ and $Tu_n \rightarrow v$, then $v = Tu^*$.

Lemma 2.3 ([27, 29]). *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{Cv}(E)$ be fixed. Then, the following are equivalent:*

- (1) C is a member of $\mathcal{SBGNR}(E)$;
- (2) C is a Bregman generalized nonexpansive retract of E ;
- (3) $\nabla g(C)$ is a member of $\mathcal{Cv}(E^*)$.

Lemma 2.4 ([29]). *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{Cv}(E)$ and $R_C^g : E \rightarrow C$ be a sunny Bregman generalized nonexpansive retraction. Then, the following hold:*

- (1) $w = R_C^g u \iff \langle u - w, \nabla g(w) - \nabla g(u) \rangle \geq 0, \forall v \in C;$
- (2) $D_g(u, R_C^g u) + D_g(R_C^g u, v) \leq D_g(u, v), \forall v \in C.$

Lemma 2.5 ([29]). *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}(E) \cap \mathcal{SBGNR}(E)$ be fixed. Then, the sunny Bregman generalized nonexpansive retraction from E onto C is uniquely determined.*

Lemma 2.6 ([29]). *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}(E)$ be such that $\nabla g(C) \in \mathcal{Cv}(E^*)$. Let $T : C \rightarrow E$ be a Bregman generalized nonexpansive mapping. Then, $F(T)$ belongs to $\mathcal{C}(E)$ and $\nabla g(F(T))$ is a member of $\mathcal{Cv}(E)$.*

Lemma 2.7 ([29]). *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}(E)$ be such that $\nabla g(C)$ is a member of $\mathcal{Cv}(E)$. Let $T : C \rightarrow E$ be a Bregman generalized nonexpansive mapping. Then, $F(T)$ is a member of $\mathcal{SBGNR}(E)$.*

Lemma 2.8 ([28]). *Let $g \in \Lambda_1(E)$ and $\{u_n\}_{n \in \mathbb{N}}, \{v_n\}_{n \in \mathbb{N}} \subset E$ be bounded. Then the following statements are equivalent:*

- (1) $\lim_{n \rightarrow \infty} D_g(u_n, v_n) = 0.$
- (2) $\lim_{n \rightarrow \infty} \|u_n - v_n\| = 0.$

Theorem 2.9 ([36]). *For a convex mapping $g \in LB(E)$, the following are equivalent:*

- (i) g is strongly coercive and belongs to $UCB(E)$;
- (ii) $\text{dom} g^* = E^*$ and g^* belongs to $LB(E^*) \cap USB(E^*)$;
- (iii) $\text{dom} g^* = E^*$ and ∇g^* is uniformly continuous on E^* .

Theorem 2.10 ([36]). *For a strongly coercive mapping $g \in \Theta_0(E)$, we have the following equivalent assertions:*

- (i) g belongs to $LB(E) \cap USB(E)$;
- (ii) ∇g^* is uniformly continuous on E^* ;
- (iii) $\text{dom} g^* = E^*$ and g^* belongs to $LB(E^*) \cap USB(E^*)$.

Lemma 2.11 ([28]). *Let $r > 0$ be a constant and $g \in UCB(E)$ be a convex function with the gauge function ρ_r . Then*

$$D_g \left(\sum_{i=0}^n \gamma_i u_i, z \right) \leq \sum_{i=0}^n \gamma_i D_g(z, u_i) - \gamma_k \gamma_l \rho_r(\|\nabla g(u_k) - \nabla g(u_l)\|),$$

for all $k, l \in \{0, 1, 2, \dots, n\}$, $u_i \in B_r$, $\gamma_i \in (0, 1)$ and $i = 0, 1, 2, \dots, n$ with $\sum_{i=0}^n \gamma_i = 1$, where ρ_r is the gauge of g .

Lemma 2.12 ([4, 36]). *Let $g \in \Lambda_1(E)$, then we have*

- (1) ∇g norm-to-weak* continuous, onto and one-to-one;
- (2) $\langle u - v, \nabla g(u) - \nabla g(v) \rangle = 0 \iff u = v$;
- (3) $\{u \in E : D_g(u, v) \leq s\}$ is bounded $\forall v \in E$ and $s > 0$;
- (4) g^* is Gâteaux differentiable, $\text{dom} g^* = E^*$ and $\nabla g^* = (\nabla g)^{-1}$.

3. FIXED POINT RESULTS FOR BREGMAN GENERALIZED NONEXPANSIVE MAPPINGS

In this section, we investigate fixed point theorems for Bregman generalized non-expansive mappings. Without using the Kadec-Klee property of a Banach space E , we extend a result due to Chang et al., [6] in the sense of Bregman distances.

Lemma 3.1. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{Cv}(E)$ be fixed. Let $\{u_n\}_{n \in \mathbb{N}}$ and $\{v_n\}_{n \in \mathbb{N}}$ be two sequences in E such that $\lim_{n \rightarrow \infty} \|u_n - u\| = 0$ and $\lim_{n \rightarrow \infty} D_g(u_n, v_n) = 0$. Then, $\lim_{n \rightarrow \infty} \|v_n - u\| = 0$.*

Proof. We first verify the boundedness of $\{v_n\}_{n \in \mathbb{N}}$. Employing (2.1), we arrive at

$$D_g(u_n, v_n) = D_g(u_n, u) + D_g(u, v_n) + \langle u_n - u, \nabla g(u) - \nabla g(v_n) \rangle, \quad \forall n \in \mathbb{N}.$$

This amounts to

$$\begin{aligned} D_g(u_n, v_n) &= D_g(u_n, u) + D_g(u, v_n) + \langle u_n - u, \nabla g(u) - \nabla g(v_n) \rangle \\ &= D_g(u_n, u) + D_g(u, v_n) - \langle u - u_n, \nabla g(u) - \nabla g(v_n) \rangle \\ &= D_{g^*}(\nabla g(u), \nabla g(u_n)) + D_{g^*}(\nabla g(v_n), \nabla g(u)) \\ &\quad - \langle u - u_n, \nabla g(u) - \nabla g(v_n) \rangle \\ &\geq D_{g^*}(\nabla g(u), \nabla g(u_n)) + D_{g^*}(\nabla g(v_n), \nabla g(u)) \\ &\quad - \|u - u_n\| \|\nabla g(u) - \nabla g(v_n)\| \\ &\geq D_{g^*}(\nabla g(u), \nabla g(u_n)) + D_{g^*}(\nabla g(v_n), \nabla g(u)) \\ &\quad - (\|u\| + \|u_n\|)(\|\nabla g(u)\| + \|\nabla g(v_n)\|). \end{aligned}$$

Thus we get

$$\begin{aligned} \frac{D_g(u_n, v_n)}{\|\nabla g(v_n)\|} &\geq \frac{D_{g^*}(\nabla g(u), \nabla g(u_n))}{\|\nabla g(v_n)\|} + \frac{D_{g^*}(\nabla g(v_n), \nabla g(u))}{\|\nabla g(v_n)\|} \\ &\quad - \frac{(\|u\| + \|u_n\|)(\|\nabla g(u)\| + \|\nabla g(v_n)\|)}{\|\nabla g(v_n)\|}. \end{aligned}$$

This, together with the coerciveness of g^* , ensures that

$$\begin{aligned} 0 &= \limsup_{n \rightarrow \infty} \frac{D_g(u_n, v_n)}{\|\nabla g(v_n)\|} \\ &\geq \limsup_{n \rightarrow \infty} \frac{D_{g^*}(\nabla g(u), \nabla g(u_n))}{\|\nabla g(v_n)\|} + \lim_{n \rightarrow \infty} \frac{D_{g^*}(\nabla g(v_n), \nabla g(u))}{\|\nabla g(v_n)\|} \\ &\quad - \liminf_{n \rightarrow \infty} \frac{(\|u\| + \|u_n\|)(\|\nabla g(u)\| + \|\nabla g(v_n)\|)}{\|\nabla g(v_n)\|} \\ &= +\infty \end{aligned}$$

which is absurd. If we apply Lemma 2.8, then we achieve the result. $\square$

Theorem 3.2. *Let $g \in \Lambda_1(E)$ be fixed. Let $T_n : E \rightarrow E$ ($n \in \mathbb{N}$) be Bregman generalized nonexpansive mappings and Ω be a set of closed and Bregman generalized-nonexpansive mappings on E such that $\bigcap_{n=1}^{\infty} F(T_n) = F(\Omega) \neq \emptyset$. Let $\{u_n\}_{n \in \mathbb{N}}$ be*

produced by:

$$(3.1) \quad \begin{cases} u_0, u_1 \in E, & A_1 = E, \\ z_n = u_n + \gamma_n(u_n - u_{n-1}), \\ x_n = \beta z_n + (1 - \beta)T_n z_n, \\ A_{n+1} = \{u \in A_n : D_g(x_n, u) \leq D_g(z_n, u)\}, \\ u_{n+1} = R_{A_{n+1}}^g u_0, \quad \forall n \in \mathbb{N}. \end{cases}$$

If $\{T_n\}_{n \in \mathbb{N}}$ enjoys the ∇g -NST-condition with Ω , $\{\gamma_n\}_{n \in \mathbb{N}} \subset (0, 1)$, and $\beta \in (0, 1)$, then, $\{u_n\}_{n \in \mathbb{N}}$ converges strongly to $R_{F(\Omega)}^g u_0$.

Proof. We discuss thorough the following several steps.

Step 1: The sequence $\{u_n\}_{n \in \mathbb{N}}$ is well-defined and $F(\Omega) \subset A_n, \forall n \geq 1$.

We show that $\nabla g(A_n)$ is a member of $\mathcal{C}v(E^*)$. Evidently, $A_1 = E$ is a member of $\mathcal{C}v(E)$. Assume that there exists $n \in \mathbb{N}$ such that $A_n \in \mathcal{C}v(E)$, then by the definition of A_{n+1} , we deduce that $A_{n+1} = \{u \in A_n : \langle z_n - x_n, \nabla g(u) \rangle \leq g(z_n) - g(x_n)\}$. Thus, $A_{n+1} \in \mathcal{C}v(E)$ and hence $\nabla g(A_n) \in \mathcal{C}v(E^*)$. By Lemma 2.3 we deduce that $A_n \in \mathcal{SBGNR}(E)$ and hence the well-definedness of $\{u_n\}_{n \in \mathbb{N}}$ is guaranteed. Next, we verify that $F(\Omega) \subset A_n, \forall n \in \mathbb{N}$. It is evident to conclude that $F(\Omega) \subset A_1$. If $F(\Omega) \subset A_n$ for some $n \in \mathbb{N}$, then, for any $p \in F(\Omega)$, we get

$$(3.2) \quad \begin{aligned} D_g(x_n, p) &= D_g(\beta z_n + (1 - \beta)T_n z_n, p) \\ &\leq \beta D_g(z_n, p) + (1 - \beta)D_g(T_n z_n, p) \\ &\leq \beta D_g(z_n, p) + (1 - \beta)D_g(z_n, p) \leq D_g(z_n, p). \end{aligned}$$

This assures that $p \in A_{n+1}$ and so $F(\Omega) \subset A_n, \forall n \in \mathbb{N}$.

Step 2: The sequence $\{u_n\}_{n \in \mathbb{N}}$ is convergent.

Let us verify that $\{u_n\}_{n \in \mathbb{N}}$ is bounded. In the light of Lemma 2.4, we obtain that

$$\begin{aligned} D_g(u_n, u_0) &= D_g(R_{A_n}^g u_0, u_0) \\ &\leq D_g(p, u_0) - D_g(p, R_{A_n}^g u_0) \leq D_g(p, u_0), \quad \forall p \in F(\Omega) \subset A_n. \end{aligned}$$

This guarantees the boundedness of sequences $\{D_g(u_n, u_0)\}_{n \in \mathbb{N}}$ and $\{u_n\}_{n \in \mathbb{N}}$. Since $u_{n+1} \in A_{n+1} \subset A_n$ and $u_n = R_{A_n}^g u_0$, we get that $D_g(u_n, u_0) \leq D_g(u_{n+1}, u_0)$, and hence $\{D_g(u_n, u_0)\}_{n \in \mathbb{N}}$ is nondecreasing which leads to the existence of $\lim_{n \rightarrow \infty} D_g(u_n, u_0)$.

The Eberlin-Smulian Theorem [16] and the boundedness of $\{u_n\}_{n \in \mathbb{N}}$ assure the existence of $\{u_{n_j}\}_{j \in \mathbb{N}} \subset \{u_n\}_{n \in \mathbb{N}}$ with $u_{n_j} \rightharpoonup q \in A_{n_j}$. In view of $u_{n_j} = R_{A_{n_j}}^g u_0$ and Lemma 2.4, we get that $D_g(u_{n_j}, u_0) \leq D_g(p, u_0), \forall j \in \mathbb{N}$. The weak lower semi-continuity of g entails to

$$D_g(q, u_0) \leq \liminf_{j \rightarrow \infty} D_g(u_{n_j}, u_0) \leq \limsup_{j \rightarrow \infty} D_g(u_{n_j}, u_0) \leq D_g(q, u_0),$$

which implies that $\lim_{j \rightarrow \infty} D_g(u_{n_j}, u_0) = D_g(q, u_0)$. Employing Lemma 3.1, we arrive at $\lim_{j \rightarrow \infty} u_{n_j} = q$. Since $\lim_{n \rightarrow \infty} D_g(u_n, u_0)$ exists and $\lim_{j \rightarrow \infty} D_g(u_{n_j}, u_0) = D_g(q, u_0)$, then $\lim_{n \rightarrow \infty} D_g(u_n, u_0) = D_g(q, u_0)$.

Next, we show that $\lim_{n \rightarrow \infty} u_n = q$. If not, then for some subsequence $\{u_{n_i}\}_{i \in \mathbb{N}}$ of $\{u_n\}_{n \in \mathbb{N}}$ with $\lim_{i \rightarrow \infty} u_{n_i} = z$ and $q \neq z$, applying Lemma 2.4, we obtain

$$\begin{aligned} D_g(q, z) &= \lim_{i, j \rightarrow \infty} D_g(u_{n_j}, u_{n_i}) = \lim_{i, j \rightarrow \infty} D_g(R_{A_{n_j}}^g u_0, R_{A_{n_i}}^g u_0) \\ &\leq D_g(u_0, R_{A_{n_i}}^g u_0) - D_g(u_0, R_{A_{n_j}}^g u_0) \\ &= \lim_{i, j \rightarrow \infty} (D_g(u_0, u_{n_j}) - D_g(u_0, u_{n_i})) = 0, \end{aligned}$$

and hence $q = z$. This is a contradiction which ensures $\lim_{n \rightarrow \infty} u_n = q$.

Step 3: $\|z_n - u_n\| \rightarrow 0$, $\|u_n - x_n\| \rightarrow 0$ and $\|x_n - z_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Since $\{u_n\}_{n \in \mathbb{N}}$ is convergent, by employing (3.1) we get $\lim_{n \rightarrow \infty} \|z_n - u_n\| = 0$. By Lemma 2.4, we arrive at $\lim_{n \rightarrow \infty} D_g(z_n, u_n) = 0$. Since $u_n \in A_n$, from (3.2), we deduce that $D_g(x_n, u_n) \leq D_g(z_n, u_n)$, and this leads to $\lim_{n \rightarrow \infty} D_g(x_n, u_n) = 0$. Hence, employing Lemma 3.1 we obtain that $\lim_{n \rightarrow \infty} \|x_n - u_n\| = 0$. Also we have $\|x_n - z_n\| \leq \|x_n - u_n\| + \|u_n - z_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Step 4: $\lim_{n \rightarrow \infty} \|z_n - Tz_n\| = 0$ and $q \in F(T)$.

Applying (3.2) we are led to

$$\begin{aligned} \|u_n - x_n\| &= \|u_n - \beta z_n - (1 - \beta)T_n z_n\| \\ &= \|(1 - \beta)(u_n - T_n z_n) - \beta(z_n - u_n)\| \\ &\leq (1 - \beta)\|u_n - T_n z_n\| - \beta\|z_n - u_n\|. \end{aligned}$$

This amounts to

$$(3.3) \quad \|u_n - T_n z_n\| \leq \frac{1}{1 - \beta} (\|u_n - x_n\| + \beta\|z_n - u_n\|) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Using (3.3) we arrive at

$$(3.4) \quad \|z_n - T_n z_n\| \leq \|z_n - u_n\| + \|u_n - T_n z_n\| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Now, the ∇g -NST-condition and (3.4) assure that

$$\lim_{k \rightarrow \infty} \|z_n - T_n z_n\| = 0, \quad \forall T \in \Omega.$$

By the relation $\lim_{n \rightarrow \infty} z_n = q$ in Step 3 and the closedness of T , we obtain $T(q) = q$.

Step 5: $\lim_{n \rightarrow \infty} u_n = R_{F(\Omega)}^g u_0$.

In the light of Lemma 2.4, we are led to

$$(3.5) \quad D_g(q, u_0) \leq D_g(R_{F(\Omega)}^g u_0, u_0).$$

Since $q \in F(\Omega) \subset A_n$ and $u_n = R_{A_n}^g u_0$, Lemma 2.4 assures that $D_g(u_n, u_0) \leq D_g(R_{F(\Omega)}^g u_0, u_0)$, consequently

$$(3.6) \quad D_g(R_{F(\Omega)}^g u_0, u_0) \leq D_g(q, u_0).$$

Combining relations (3.5) and (3.6), we obtain that $D_g(R_{F(\Omega)}^g u_0, u_0) = D_g(q, u_0)$ which ensures that $q = R_{F(\Omega)}^g u_0$. $\square$

Remark 3.3. Theorem 3.2 improves Theorems 1.1 and 1.3 in the following aspects.

- (1) For the structure of Banach spaces, we extend the duality mapping to a more general case, that is, a convex and Gâteaux differentiable function.
- (2) For the mappings, we extend the mapping from a generalized nonexpansive mapping to Bregman generalized ∇g -nonexpansive mappings. We remove the nonexpansivity assumption on the mapping T and extend the result to a countable family of Bregman generalized nonexpansive mappings.
- (3) For the algorithm, we remove the sets C_n and Q_n in Theorem 1.1 and replaced them with the set A_n .
- (4) For the spaces, we extend the results from a uniformly smooth and uniformly convex real Banach space to a reflexive Banach space.

4. ∇g -FIXED POINT RESULTS

In this section, beyond the usual notion of fixed points, we first introduce a new notion of ∇g -fixed points for the mapping $S : E \rightarrow E^*$ and then investigate strong convergence theorems for these mappings.

Definition 4.1. Let $g \in \Lambda_1(E)$ and $S : E \rightarrow E^*$ be fixed mappings. A point $p \in E$ is called a ∇g -fixed point of S if $S(p) = \nabla g(p)$, where ∇g is the gradient mapping on E . The set of ∇g -fixed points of S is denoted by $F_{\nabla g}(S)$.

Definition 4.2. Let $g \in \Lambda_1(E)$ be fixed. A mapping $S : E \rightarrow E^*$ is called ∇g -nonexpansive if:

$$\|Su - Sv\| \|u - v\| \leq \langle u - v, \nabla g(u) - \nabla g(v) \rangle, \quad \forall u, v \in E.$$

Definition 4.3. Let $C \subset E$ be fixed. A mapping $S : C \rightarrow E^*$ is called *Bregman generalized- ∇g -nonexpansive* if $F_{\nabla g}(S) \neq \emptyset$ and $D_g((\nabla g^* \circ S)p, v) \leq D_g(p, v)$, $\forall p \in F_{\nabla g}(S)$, $v \in C$.

∇g -NST-Condition. Let $g \in \Lambda_1(E)$ be fixed. Assume that $\{S_n\}_{n \in \mathbb{N}}$ and Ω be two families of Bregman generalized- ∇g -nonexpansive mappings from C into E^* such that $\bigcap_{n=1}^{\infty} F(S_n) = F(\Omega) \neq \emptyset$, where $F_{\nabla g}(S_n)$ is the set of ∇g -fixed points of S_n and $F_{\nabla g}(\Omega)$ is the set of ∇g -fixed points of Ω . Then, $\{S_n\}_{n \in \mathbb{N}}$ enjoys the ∇g -NST-condition with Ω if for each bounded sequence $\{u_n\}_{n \in \mathbb{N}} \subset C$,

$$\lim_{n \rightarrow \infty} \|\nabla g(u_n) - S_n(u_n)\| = 0 \Rightarrow \lim_{n \rightarrow \infty} \|\nabla g(u_n) - S(u_n)\| = 0, \quad \forall S \in \Omega.$$

Definition 4.4. Let $g \in \Lambda_1(E)$ be fixed. A mapping $S : E \rightarrow E^*$ is called ∇g -pseudocontractive if

$$\langle S(u) - S(v), u - v \rangle \leq \langle u - v, \nabla g(u) - \nabla g(v) \rangle, \quad \forall u, v \in E.$$

Definition 4.5. We say that $S : E \rightarrow E^*$ is ∇g -nonexpansive if

$$\|S(u) - S(v)\| \|u - v\| \leq \langle u - v, \nabla g(u) - \nabla g(v) \rangle, \quad \forall u, v \in E.$$

Remark 4.6. (i) We observe that if a mapping $S : E \rightarrow E^*$ is ∇g -nonexpansive then S is ∇g -pseudocontractive. Indeed, we have

$$\langle S(u) - S(v), u - v \rangle \leq \|S(u) - S(v)\| \|u - v\| \leq \langle u - v, \nabla g(u) - \nabla g(v) \rangle, \quad \forall u, v \in E,$$

and satisfies the inequality

$$\|S(u) - S(v)\| \leq \|\nabla g(u) - \nabla g(v)\|, \quad \forall u, v \in E.$$

(ii) If a mapping $S : E \rightarrow E^*$ is ∇g -pseudocontractive, then the mapping $\Phi := (\nabla g - S)$ is monotone. Indeed,

$$\langle \Phi u - \Phi v, u - v \rangle = \langle \nabla g(u) - \nabla g(v), u - v \rangle - \langle Su - Sv, u - v \rangle \geq 0, \quad \forall u, v \in E.$$

Theorem 4.7. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{Cv}(E)$ be such that $\nabla g(C)$ belongs to $\mathcal{Cv}(E^*)$. Let $S_n : E \rightarrow E^*$, $n \in \mathbb{N}$ be a countable family of generalized- ∇g -nonexpansive mappings and Ω be a family of closed and Bregman generalized- ∇g -nonexpansive mappings from E to E^* such that $\bigcap_{n=1}^{\infty} F_{\nabla g}(S_n) = F_{\nabla g}(\Omega) \neq \emptyset$. Assume that $\{u_n\}_{n \in \mathbb{N}}$ be defined by:*

$$(4.1) \quad \begin{cases} u_0, u_1 \in E, & A_1 = E, \\ z_n = u_n + \gamma_n(u_n - u_{n-1}), \\ u_n = \beta z_n + (1 - \beta)(\nabla g^* o S_n)z_n, \\ A_{n+1} = \{u \in A_n : D_g(u_n, u) \leq D_g(z_n, u)\}, \\ u_{n+1} = R_{A_{n+1}}^g u_0, \quad \forall n \in \mathbb{N}. \end{cases}$$

Let $\{T_n\}_{n \in \mathbb{N}}$ satisfy the ∇g -NST-condition with Ω , $\{\gamma_n\}_{n \in \mathbb{N}} \subset (0, 1)$ and $\beta \in (0, 1)$. Then, $\{u_n\}_{n \in \mathbb{N}}$ converges strongly to $R_{F_{\nabla g}(\Omega)}^g u_0$.

Proof. Let $H_n : E \rightarrow E$, $n \in \mathbb{N}$ be defined by $H_n := (\nabla g^* o S_n)$. Then, H_n is Bregman generalised nonexpansive and $\bigcap_{n=1}^{\infty} F_{\nabla g}(H_n) = \bigcap_{n=1}^{\infty} F_{\nabla g}(S_n) = F(\Omega)$. In the light of Theorem 3.2, the sequence $\{u_n\}_{n \in \mathbb{N}}$ defined in (4.1) is strongly convergent to a point $q \in R_{F_{\nabla g}(\Omega)}^g u_0$. $\square$

Lemma 4.8 ([15], Corollary 2.7). *Let $\Phi : E \rightarrow E^*$ be a semi-continuous and monotone mapping. Then Φ is maximally monotone.*

The following result extends the corresponding one of Bruck [3] in the sense of Bregman distances and it will be central in the sequel.

Theorem 4.9. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{Cv}(E)$ be fixed. Let $S_n : C \rightarrow E^*$, $n \in \mathbb{N}$ be a family of ∇g -nonexpansive mappings such that $\bigcap_{n=1}^{\infty} F_{\nabla g}(S_n) \neq \emptyset$. Let the mapping $S : C \rightarrow E^*$ be defined by $S(x) = \sum_{n=1}^{\infty} \eta_n S_n(x)$, where $\{\eta_n\}_{n \in \mathbb{N}} \subset (0, 1)$ and $\sum_{n=1}^{\infty} \eta_n = 1$. Then*

- (1) S is ∇g -nonexpansive;
- (2) $F_{\nabla g}(S) = \bigcap_{n=1}^{\infty} F_{\nabla g}(S_n)$;
- (3) $(\nabla g - S)$ is maximal monotone.

Proof. By the inequalities

$$\|S_n(v)\| \leq \|S_n(v) - \nabla g(u_0)\| + \|\nabla g(u_0)\| \leq \|\nabla g(v) - \nabla g(u_0)\| + \|\nabla g(u_0)\|,$$

we deduce that S is well-defined. This verifies that $\sum_{n=1}^{\infty} \eta_n S_n(v)$ converges strongly in C .

(1) Using the ∇g -nonexpansiveness of S_n we obtain that

$$\|S(v) - S(u)\| \|v - u\| = \left\| \sum_{n=1}^{\infty} \eta_n S_n(v) - \sum_{n=1}^{\infty} \eta_n S_n(u) \right\| \|v - u\|$$

$$\begin{aligned}
&= \left\| \sum_{n=1}^{\infty} \eta_n (S_n(v) - S_n(u)) \right\| \|v - u\| \\
&\leq \sum_{n=1}^{\infty} \eta_n \|S_n(v) - S_n(u)\| \|v - u\| \\
&\leq \sum_{n=1}^{\infty} \eta_n \langle v - u, \nabla g(v) - \nabla g(u) \rangle \\
&\leq \langle v - u, \nabla g(v) - \nabla g(u) \rangle
\end{aligned}$$

and hence S is ∇g -nonexpansive.

(2) Obviously we have $\cap_{n=1}^{\infty} F_{\nabla g}(S_n) \subset F_{\nabla g}(S)$. Conversely, let $u_0 \in \cap_{n=1}^{\infty} F_{\nabla g}(S_n)$ and $v \in F_{\nabla g}(S)$. Then,

$$\begin{aligned}
\|\nabla g(v) - \nabla g(u_0)\| &= \|S(v) - \nabla g(u_0)\| = \left\| \sum_{n=1}^{\infty} \eta_n S_n(v) - \nabla g(u_0) \right\| \\
(4.2) \quad &= \left\| \sum_{n=1}^{\infty} \eta_n (S_n(v) - \nabla g(u_0)) \right\| \leq \sum_{n=1}^{\infty} \eta_n \|S_n(v) - S_n(u)\| \\
&\leq \sum_{n=1}^{\infty} \eta_n \|S_n(v) - \nabla g(u_0)\|.
\end{aligned}$$

Also, we have $S_n(u_0) = \nabla g(u_0)$ and S_n is ∇g -nonexpansive. In view of (4.2), we get that $\|S_n(v) - S_n(u_0)\| \leq \|\nabla g(v) - \nabla g(u_0)\|$. From (4.1) and (4.2) we see that,

$$(4.3) \quad \left\| \sum_{n=1}^{\infty} \eta_n (S_n(v) - \nabla g(u_0)) \right\| = \|\nabla g(v) - \nabla g(u_0)\|$$

and

$$(4.4) \quad \|S_n(v) - \nabla g(u_0)\| = \|\nabla g(v) - \nabla g(u_0)\|.$$

The strict convexity of E^* and the assumption $\sum_{n=1}^{\infty} \eta_n = 1$ ($\eta_n > 0$, $\forall n \in \mathbb{N}$) incorporating with (4.3) and (4.4) imply that $S_n(v) - \nabla g(u_0) = S_k(v) - \nabla g(u_0)$ $\forall n, k \in \mathbb{N}$ and hence $S_n(v) = S_k(v)$ $\forall n, k \in \mathbb{N}$. This amounts to

$$\nabla g(v) = S(v) = \sum_{n=1}^{\infty} \eta_n S_n(v) = \sum_{k=1}^{\infty} \eta_k S_k(v), \quad \forall k \in \mathbb{N},$$

which leads to $v \in \cap_{n=1}^{\infty} F_{\nabla g}(S_n)$.

(3) Due to the fact that S is ∇g -nonexpansive, in the light of Remark 4.6 we deduce that it is ∇g -pseudocontractive and hence, by Remark 4.6, $(\nabla g - S)$ is monotone. It is evident that $(\nabla g - S)$ has a full domain E and is continuous. Employing Lemma 4.8, we deduce that $(\nabla g - S)$ is maximal monotone. $\square$

5. ∇ -GENERALIZED MIXED EQUILIBRIUM PROBLEMS

This section devotes to the ∇g -generalized mixed equilibrium problem in Banach spaces. We first mention the following remark.

Remark 5.1. Let $C \in \mathcal{C}(E)$ be such that $\nabla g(C)$ is closed and convex. Let $\psi : \nabla g(C) \rightarrow \mathbb{R}$ be a lower semi-continuous and convex function and $\Phi : C \rightarrow E^*$ be a continuous and monotone mapping. For solving the ∇g -generalized mixed equilibrium problems, we assume that the bi-function $h : \nabla g(C) \times \nabla g(C) \rightarrow \mathbb{R}$ satisfies the following conditions:

- (C1) $h(x^*, x^*) = 0$ for all $x^* \in \nabla g(C)$;
- (C2) h is monotone, i.e. $h(x^*, y^*) + h(y^*, x^*) \leq 0$ for all $x^*, y^* \in \nabla g(C)$;
- (C3) $\limsup_{t \downarrow 0} h(x^* + t(z^* - x^*), y^*) \leq h(x^*, y^*)$ for all $x^*, y^*, z^* \in \nabla g(C)$;
- (C4) For all $x^* \in \nabla g(C)$, the function $h(x^*, \cdot)$ is convex and lower semi-continuous.

We present the following new lemmas which will be essential in the sequel.

Lemma 5.2. Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}v(E)$ be such that $\nabla g(C)$ is closed and convex. Let $\psi : \nabla g(C) \rightarrow \mathbb{R}$ be a lower semi-continuous and convex function. Let $\Phi : C \rightarrow E^*$ be a continuous and monotone mapping. Assume $f : \nabla g(C) \times \nabla g(C) \rightarrow \mathbb{R}$ is a bi-function. For any $r > 0$ and $x \in E$, we define a mapping $T_r : E \rightarrow C$ by

$$T_r(x) = \left\{ u \in C : h(\nabla g(u), \nabla g(z)) + \psi(\nabla g(z)) - \psi(\nabla g(u)) \right. \\ \left. + \langle \Phi u, z - u \rangle + \frac{1}{r} \langle u - x, \nabla g(z) - \nabla g(u) \rangle \geq 0, \forall z \in C \right\}.$$

Then, the function T_r fulfills the following properties:

- (i) T_r is single-valued,
- (ii) T_r is a Bregman firmly nonexpansive mapping, that is,
 $\langle T_r x - T_r y, \nabla g(T_r x) - \nabla g(T_r y) \rangle \leq \langle x - y, \nabla g(T_r x) - \nabla g(T_r y) \rangle, \forall x, y \in E,$
- (iii) $F(T_r) = \text{BGMEP}(h, \Phi, \psi)$,
- (iv) $\text{BGMEP}(h, \Phi, \psi) \in \mathcal{C}(E)$ and $\nabla g(\text{BGMEP}(h, \Phi, \psi)) \in \mathcal{C}v(E^*)$,
- (v) $D_g(x, T_r x) + D_g(T_r x, q) \leq D_g(x, q)$ for all $q \in F(T_r)$ and $x \in E$.

Proof. (i) Let $x \in E$, $r > 0$ and $u_1, u_2 \in T_r(x)$ be fixed. Then, we have

$$h(\nabla g(u_1), \nabla g(u_2)) + \psi(\nabla g(u_2)) - \psi(\nabla g(u_1)) \\ + \langle \Phi u_1, u_2 - u_1 \rangle + \frac{1}{r} \langle u_1 - x, \nabla g(u_2) - \nabla g(u_1) \rangle \geq 0, \\ h(\nabla g(u_2), \nabla g(u_1)) + \psi(\nabla g(u_1)) - \psi(\nabla g(u_2)) \\ + \langle \Phi u_2, u_1 - u_2 \rangle + \frac{1}{r} \langle u_2 - x, \nabla g(u_1) - \nabla g(u_2) \rangle \geq 0.$$

These together with condition (C2) and the monotonicity of Φ lead to

$$\frac{1}{r} \langle u_1 - u_2, \nabla g(u_2) - \nabla g(u_1) \rangle \geq 0.$$

From monotonicity of ∇g and the strict convexity of E , we conclude that $u_1 = u_2$. which implies that T_r is single-valued.

(ii) For any $x, y \in C$, we obtain

$$h(\nabla g(T_r x), \nabla g(T_r y)) + \psi(\nabla g(T_r y)) - \psi(\nabla g(T_r x)) + \langle \Phi T_r x, T_r y - T_r x \rangle \\ + \frac{1}{r} \langle T_r x - x, \nabla g(T_r y) - \nabla g(T_r x) \rangle \geq 0,$$

$$h(\nabla g(T_r y), \nabla g(T_r x)) + \psi(\nabla g(T_r x)) - \psi(\nabla g(T_r y)) + \langle \Phi T_r y, T_r x - T_r y \rangle \\ + \frac{1}{r} \langle T_r y - y, \nabla g(T_r x) - \nabla g(T_r y) \rangle \geq 0.$$

This, together with condition (C2) and the monotonicity of Φ , implies that

$$\langle T_r x - T_r y, \nabla g(T_r x) - \nabla g(T_r y) \rangle \leq \langle x - y, \nabla g(T_r x) - \nabla g(T_r y) \rangle.$$

(iii) We prove that $F(T_r) = \text{BGMEP}(h, \psi, \Phi)$.

$$u = T_r u \iff h(\nabla g(u), \nabla g(z)) + \psi(\nabla g(z)) - \psi(\nabla g(u)) \\ + \langle \Phi u, z - u \rangle + \frac{1}{r} \langle u - u, \nabla g(z) - \nabla g(u) \rangle \geq 0, \quad \forall z \in C \\ \iff h(\nabla g(u), \nabla g(z)) + \psi(\nabla g(z)) - \psi(\nabla g(u)) + \langle \Phi u, z - u \rangle \geq 0, \quad \forall z \in C, \\ \iff u \in \text{BGMEP}(h, \psi, \Phi).$$

(iv) $\text{BGMEP}(h, \psi, \Phi) \in \mathcal{C}(E)$ and $\nabla g(\text{BGMEP}(h, \psi, \Phi)) \in \mathcal{C}v(E^*)$.

Clearly, $\text{BGMEP}(h, \psi, \Phi)$ is closed. Let $\{u_n^*\}_{n \in \mathbb{N}} \subset \nabla g(\text{BGMEP}(h, \psi, \Phi))$ such that $u_n^* \rightarrow u^*$ for some $u^* \in E^*$. Since $\nabla g(C)$ is closed, $u^* \in \nabla g(C)$. Hence, there exist $u \in C$ and $\{u_n\}_{n \in \mathbb{N}} \subset \text{BGMEP}(h, \psi, \Phi)$ such that $u^* = \nabla g(u)$ and $u_n^* = \nabla g(u_n)$, $\forall n \in \mathbb{N}$.

Utilizing the properties of h, ψ, Φ and the fact that ∇g^* is uniformly continuous on bounded subsets of E^* , we obtain that

$$\psi(u^*) \leq \liminf_{n \rightarrow \infty} \psi(u_n^*) \\ \leq \liminf_{n \rightarrow \infty} [h(u_n^*, \nabla g(y)) + \psi(\nabla g(y)) + \langle \Phi \nabla g^*(u_n^*), y - \nabla g^*(u_n^*) \rangle] \\ \leq \limsup_{n \rightarrow \infty} [h(u_n^*, \nabla g(y)) + \psi(\nabla g(y)) + \langle \Phi \nabla g^*(u_n^*), y - \nabla g^*(u_n^*) \rangle] \\ \leq h(u^*, \nabla g(y)) + \psi(\nabla g(y)) + \langle \Phi \nabla g^*(u^*), y - \nabla g^*(u^*) \rangle.$$

Hence, $\nabla g(\text{BGMEP}(h, \psi, \Phi))$ is closed.

Now, let $u_1^*, u_2^* \in \nabla g(\text{BGMEP}(h, \psi, \Phi))$. Then, $u_1^* = \nabla g(u_1)$, $u_2^* = \nabla g(u_2)$ for some $u_1, u_2 \in C$. For $\lambda \in (0, 1]$, let $u_\lambda^* = \lambda u_1^* + (1 - \lambda)u_2^* \in \nabla g(C)$. For any $y \in C$, set $z_t^* = t \nabla g(y) + (1 - t)u_\lambda^*$.

The conditions (C1)-(C4) assure that:

$$0 = h(z_t^*, z_t^*) + \psi(z_t^*) - \psi(z_t^*) + \langle \Phi(\nabla g^{-1}(z_t^*)), y - \nabla g^*(z_t^*) \rangle \\ - \langle \Phi(\nabla g^*(z_t^*)), y - \nabla g^*(z_t^*) \rangle \\ \leq h(z_t^*, \nabla g(y)) + \psi(\nabla g(y)) - \psi(z_t^*) + \langle \Phi(\nabla g^*(z_t^*)), y - \nabla g^*(z_t^*) \rangle \\ = h(u_\lambda^* + t(\nabla g(y) - u_\lambda^*), \nabla g(y)) + \psi(\nabla g(y)) - \psi(u_\lambda^* + t(\nabla g(y) - u_\lambda^*)) \\ + \langle \Phi(\nabla g^*(u_\lambda^* + t(\nabla g(y) - u_\lambda^*))), y - \nabla g^*(u_\lambda^* + t(\nabla g(y) - u_\lambda^*)) \rangle.$$

Applying condition (C3), we conclude that

$$h(u_\lambda^*, \nabla g(y)) + \psi(\nabla g(y)) - \psi(u_\lambda^*) + \langle \Phi(\nabla g^*(u_\lambda^*)), y - \nabla g^*(u_\lambda^*) \rangle \geq 0.$$

Hence, $u_\lambda^* \in \nabla g(\text{BGMEP}(h, \psi, \Phi))$. Therefore, $\nabla g(\text{BGMEP}(h, \psi, \Phi))$ is convex.

(v) We show that $D_g(x, T_r x) + D_g(T_r x, q) \leq D_g(x, q)$, $\forall q \in F(T_r)$, $x \in E$. Taking $x, y \in C$, we obtain that

$$(5.1) \quad D_g(T_r x, T_r y) + D_g(T_r y, T_r x) = \langle T_r x - T_r y, \nabla g(T_r x) - \nabla g(T_r y) \rangle$$

and

$$(5.2) \quad \begin{aligned} D_g(x, T_r y) + D_g(y, T_r x) - D_g(x, T_r x) - D_g(y, T_r y) \\ = \langle x - y, \nabla g(T_r x) - \nabla g(T_r y) \rangle. \end{aligned}$$

Applying Lemma 5.2(ii), (5.1) and (5.2), one obtains that

$$(5.3) \quad \begin{aligned} D_g(T_r x, T_r y) + D_g(T_r y, T_r x) \\ \leq D_g(x, T_r y) + D_g(y, T_r x) - D_g(x, T_r x) - D_g(y, T_r y), \quad \forall x, y \in C. \end{aligned}$$

Setting $y = u \in F(T_r)$ in (5.3), we get

$$D_g(T_r x, u) + D_g(u, T_r x) \leq D_g(x, u) + D_g(u, T_r x) - D_g(x, T_r x) - D_g(u, u), \quad \forall x \in C.$$

Thus we have

$$D_g(y, T_r x) + D_g(T_r x, u) \leq D_g(x, u), \quad \forall x \in C, u \in F(T_r).$$

□

Lemma 5.3. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}(E)$ be such that $\nabla g(C) \in \mathcal{C}v(E^*)$. Let T be a Bregman generalized ∇g -nonexpansive mapping from C to E^* with $F_{\nabla g}(T) \neq \emptyset$. Then, $F_{\nabla g}(T) \in \mathcal{C}(E)$ and $\nabla g(F_{\nabla g}(T)) \in \mathcal{C}v(E^*)$.*

Proof. First, we prove that $\nabla g(F_{\nabla g}(T))$ is convex. Let $u^*, v^* \in \nabla g(F_{\nabla g}(T))$ and $\alpha, \beta \in (0, 1)$ with $\alpha + \beta = 1$. Then, there exist $u, v \in E$ with $\nabla g(u) = u^*$ and $\nabla g(v) = v^*$ and hence we have

$$\begin{aligned} & D_g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), \nabla g^*(\alpha u^* + \beta v^*)) \\ &= g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(\nabla g^*(\alpha u^* + \beta v^*)) \\ &\quad - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*) - \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle \\ &= g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(\nabla g^*(\alpha u^* + \beta v^*)) \\ &\quad - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*) - \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle \\ &= g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(\nabla g^*(\alpha u^* + \beta v^*)) \\ &\quad - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle + \langle \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle \\ &= g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(\nabla g^*(\alpha u^* + \beta v^*)) \\ &\quad - \alpha \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), u^* \rangle - \beta \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), v^* \rangle \\ &\quad + g(\nabla g^*(\alpha u^* + \beta v^*)) + g^*(\alpha u^* + \beta v^*) - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle \\ &\quad + \langle \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle \\ &= \alpha [g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(u) - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*) - u, \nabla g(u) \rangle] \\ &\quad + \alpha [g(u) - \langle u, \nabla g(u) \rangle] + \beta [g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*)) - g(v) \\ &\quad - \langle (\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*) - v, \nabla g(v) \rangle] + \beta [g(v) - \langle v, \nabla g(v) \rangle] + g^*(\alpha u^* + \beta v^*) \\ &= \alpha D_g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), u) + \beta D_g((\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*), v) \\ &\quad + \alpha [g(u) - \langle u, \nabla g(u) \rangle] + \beta [g(v) - \langle v, \nabla g(v) \rangle] + g^*(\alpha u^* + \beta v^*) \\ &= \alpha D_g(\nabla g^*(\alpha u^* + \beta v^*), u) + \beta D_g(\nabla g^*(\alpha u^* + \beta v^*), v) + \alpha [g(u) - \langle u, \nabla g(u) \rangle] \\ &\quad + \beta [g(v) - \langle v, \nabla g(v) \rangle] + g^*(\alpha u^* + \beta v^*) \end{aligned}$$

$$\begin{aligned}
&= \alpha[g(\nabla g^*(\alpha u^* + \beta v^*)) - g(u) - \langle \nabla g^*(\alpha u^* + \beta v^*) - u, \nabla g(u) \rangle] \\
&\quad + \alpha[g(u) - \langle u, \nabla g(u) \rangle] \\
&\quad + \beta[g(\nabla g^*(\alpha u^* + \beta v^*)) - g(v) - \langle \nabla g^*(\alpha u^* + \beta v^*) - v, \nabla g(v) \rangle] \\
&\quad + \beta[g(v) - \langle v, \nabla g(v) \rangle] + g^*(\alpha u^* + \beta v^*) \\
&= g(\nabla g^*(\alpha u^* + \beta v^*)) - \langle \nabla g^*(\alpha u^* + \beta v^*), \alpha u^* + \beta v^* \rangle + g^*(\alpha u^* + \beta v^*) \\
&= 0.
\end{aligned}$$

By strict convexity of g , we get that $(\nabla g^* o T) \nabla g^*(\alpha u^* + \beta v^*) = \nabla g^*(\alpha u^* + \beta v^*)$, which ensures that $\nabla g^*(\alpha u^* + \beta v^*) \in F_{\nabla g}(T)$. Thus, $\alpha u^* + \beta v^* \in \nabla g(F_{\nabla g}(T))$, which shows that $\nabla g(F_{\nabla g}(T))$ is convex.

Let us verify that $F_{\nabla g}(T)$ and $\nabla g(F_{\nabla g}(T))$ are closed. It is evident that $F_{\nabla g}(T)$ is closed. Let $\{v_n^*\}_{n \in \mathbb{N}} \subset \nabla g(F_{\nabla g}(T))$ such that $v_n^* \rightarrow v^*$ for some $v^* \in E^*$. Since $\nabla g(C)$ is closed, we have that $v^* \in \nabla g(C)$. Hence, there exist $v \in C$ and $\{v_n\}_{n \in \mathbb{N}} \subset F_{\nabla g}(T)$ such that $v^* = \nabla g(v)$ and $v_n^* = \nabla g(v_n)$, $\forall n \in \mathbb{N}$. Thus we obtain

$$D_g((\nabla g^* o T)v, v) = \lim_{n \rightarrow \infty} D_g((\nabla g^* o T)v, v_n) \leq \lim_{n \rightarrow \infty} D_g(v, v_n) = D_g(v, v) = 0.$$

By strict convexity of g , we deduce that $v^* = \nabla g(v) \in \nabla g(F_{\nabla g}(T))$ which shows that $\nabla g(F_{\nabla g}(T))$ is closed. $\square$

The following Lemma is deduced by Lemmas 5.2 and 5.3.

Lemma 5.4. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}(E)$ be such that $\nabla g(C) \in \mathcal{C}v(E^*)$. Let $T : C \rightarrow E^*$ be a Bregman generalized- ∇g -nonexpansive mapping such that $F_{\nabla g}(T) \neq \emptyset$. Then, $F_{\nabla g}(T)$ is a sunny Bregman generalized- ∇g -nonexpansive retract of E .*

Remark 5.5. In view of Lemmas 5.3 and 5.4, we are led to $\nabla g(F_{\nabla g}(T)) \in \mathcal{C}(E^*)$ and $\nabla g(BGMEP) \in \mathcal{C}v(E^*)$. By the strict convexity of g we deduce that

$$\nabla g(F_{\nabla g}(T) \cap BGMEP(h, \psi, \Phi)) = \nabla g(F_{\nabla g}(T)) \cap \nabla g(BGMEP(h, \psi, \Phi)).$$

By Lemma 5.3, we obtain that $F_{\nabla g}(T) \cap BGMEP(h, \psi, \Phi)$ is a sunny Bregman generalized- ∇g -nonexpansive retract of E .

We end this section by proving the following theorem in Banach spaces.

Theorem 5.6. *Let $g \in \Lambda_1(E)$ and $C \in \mathcal{C}v(E)$ be such that $\nabla g(C) \in \mathcal{C}v(E^*)$. Let $\psi : \nabla g(C) \rightarrow \mathbb{R}$ be a lower semi-continuous and convex function. Let $\Phi : C \rightarrow E^*$ be a continuous and monotone mapping. Let $f : \nabla g(C) \times \nabla g(C) \rightarrow \mathbb{R}$ be a bifunction satisfying conditions (C1)–(C4). Let $T_n : C \rightarrow E^*$ be a countable family of Bregman generalized- ∇g -nonexpansive mappings and Θ be a family of closed and Bregman generalized- ∇g -nonexpansive mappings from C to E^* such that*

$$\bigcap_{n=1}^{\infty} F_{\nabla g}(T_n) \cap BGMEP(h, A, \psi) = F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \neq \emptyset.$$

Let $\alpha \in (0, 1)$ and $\{r_n\}_{n \in \mathbb{N}} \subset [a, \infty)$ for some $a > 0$. Let $\{x_n\}_{n \in \mathbb{N}}$ be defined by

$$\begin{cases} x_1 = x \in C, & C_1 = C, \\ y_n = \alpha x_n + (1 - \alpha) \nabla g^* o T_n x_n, & v_n = K_{r_n} y_n, \\ C_{n+1} = \{v \in C_n : D_g(v_n, v) \leq D_g(x_n, v)\}, \\ x_{n+1} = R_{C_{n+1}}^g x, & n \in \mathbb{N}. \end{cases}$$

Assume that $\{T_n\}_{n \in \mathbb{N}}$ satisfies the ∇g -NST-condition with Θ , then $\{x_n\}_{n \in \mathbb{N}}$ converges strongly to $R_{F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi)}^g x$, where this operator is the sunny Bregman generalized- ∇g -nonexpansive retraction of E onto $F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi)$.

Proof. Step 1: We first notice that the sequence $\{x_n\}_{n \in \mathbb{N}}$ is well-defined and $F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_n$. Now, we show that $\nabla g(C_n) \in \mathcal{Cv}(E^*)$. Clearly, $\nabla g(C_1) = \nabla g(C) \in \mathcal{Cv}(E^*)$.

Assume that there exists $n \in \mathbb{N}$ such that $\nabla g(C_n) \in \nabla g(\mathcal{Cv}(E))$. From the definition of C_{n+1} , it is easy to see that

$$C_{n+1} = \{v \in C_n : \langle x_n - v_n, \nabla g(v) \rangle \leq g(x_n) - g(v_n)\}.$$

Thus, $\nabla g(C_{n+1}) \in \mathcal{Cv}(E^*)$, for each $n \in \mathbb{N}$ and $\nabla g(C_n) \in \mathcal{Cv}(E^*)$. By Lemma 2.4, C_n is a sunny Bregman generalized- ∇g -nonexpansive retract of E . Hence, the sequence $\{x_n\}_{n \in \mathbb{N}}$ is well-defined.

Next, we prove that

$$F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_n, \quad \forall n \in \mathbb{N}.$$

Clearly, $F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_1 = C$. Assume that

$$F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_n$$

for some $n \in \mathbb{N}$. Let $p \in F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi)$. By Lemmas 2.11 and 5.2(v) we deduce that

$$\begin{aligned} D_g(v_n, p) &= D_g(K_{r_n} y_n, p) \leq D_g(y_n, p) \\ (5.4) \quad &\leq D_g(x_n, p) - \alpha(1 - \alpha) \rho_r(\|x_n - \nabla g^* o T_n(x_n)\|) \\ &\leq D_g(x_n, p). \end{aligned}$$

Which implies that $p \in C_{n+1}$. Hence, $F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_n, \forall n \geq 1$.

Step 2: We show that $\lim_{n \rightarrow \infty} \|x_n - v_n\| = \lim_{n \rightarrow \infty} \|x_n - y_n\| = \lim_{n \rightarrow \infty} \|v_n - y_n\| = 0$.

First, we prove that the sequence $\{x_n\}_{n \in \mathbb{N}}$ is bounded. From Lemma 5.4, we conclude that $(\forall p \in F_{\nabla g}(\Theta) \cap BGMEP(h, \psi, \Phi) \subseteq C_n)$

$$D_g(x, x_n) = D_g(x, R_{C_n}^g x) \leq D_g(x, p) - D_g(R_{C_n}^g x, p) \leq D_g(x, p).$$

This implies that the sequence $\{D_g(x, x_n)\}_{n \in \mathbb{N}}$ is bounded and hence $\{x_n\}_{n \in \mathbb{N}}$ is bounded too. Since

$$x_{n+1} = R_{C_{n+1}}^g x \in C_{n+1} \subseteq C_n \quad \text{and} \quad x_n = R_{C_n}^g x,$$

we get that $D_g(x, x_n) \leq D_g(x, x_{n+1})$. This assures that $\{D_g(x, x_n)\}_{n \in \mathbb{N}}$ is nondecreasing and hence $\lim_{n \rightarrow \infty} D_g(x, x_n)$ exists.

Also, from Lemma 5.4 and the relation $x_n = R_{C_n}^g x$, we get that

$$\begin{aligned} D_g(x_n, x_m) &= D_g(R_{C_n}^g x, R_{C_m}^g x) \leq D_g(x, R_{C_m}^g x) - D_g(x, R_{C_n}^g x) \\ &= D_g(x, x_m) - D_g(x, x_n) \rightarrow 0 \quad \text{as } m, n \rightarrow \infty. \end{aligned}$$

Therefore, $\lim_{m, n \rightarrow \infty} D_g(x_n, x_m) = 0$. By Lemma 2.8, we obtain that $\lim_{m, n \rightarrow \infty} \|x_n - x_m\| = 0$ and hence $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence. Thus, there exists $x^* \in C$ such that $\lim_{n \rightarrow \infty} x_n = x^*$. In view of (5.4) we see that $D_g(v_n, x_m) \leq D_g(x_n, x_m) \rightarrow 0$ as $n \rightarrow \infty$. In the light of Lemma 2.8, we get

$$(5.5) \quad \lim_{m, n \rightarrow \infty} \|v_n - x_m\| = 0, \text{ hence } \lim_{n \rightarrow \infty} \|v_n - x_n\| = 0, \quad \lim_{n \rightarrow \infty} D_g(x_n, v_n) = 0,$$

and hence $\lim_{n \rightarrow \infty} v_n = x^*$. In the light of, Lemma 5.2(v), (5.4) and (5.5), we get that

$$\begin{aligned} D_g(y_n, v_n) &= D_g(y_n, K_{r_n} y_n) \leq D_g(y_n, p) - D_g(v_n, p) \\ &\leq D_g(x_n, p) - D_g(v_n, p) \\ &= D_g(x_n, v_n) + D_g(v_n, p) - D_g(v_n, p) + \langle x_n - v_n, \nabla g(v_n) - \nabla g(p) \rangle \\ &\leq D_g(x_n, v_n) + (\|x_n - v_n\|)(\|\nabla g(v_n)\| + \|\nabla g(p)\|) \\ &\leq D_g(x_n, v_n) + M\|x_n - v_n\| \rightarrow 0 \text{ as } n \rightarrow \infty, \end{aligned}$$

where $M = \sup\{\|\nabla g(v_n)\| + \|\nabla g(p)\| : n \in \mathbb{N}\} < \infty$. Thus, by Lemma 3.1 we conclude that $\lim_{n \rightarrow \infty} \|v_n - y_n\| = 0$. This, together with (5.5), yields to

$$\lim_{n \rightarrow \infty} \|x_n - v_n\| = \lim_{n \rightarrow \infty} \|v_n - y_n\| = \lim_{n \rightarrow \infty} \|x_n - y_n\| = 0.$$

Step 3: We show that $\lim_{n \rightarrow \infty} \|\nabla g(x_n) - Tx_n\| = 0$ for all $T \in \Theta$.

By (5.4), we obtain that

$$\begin{aligned} \alpha(1 - \alpha)\rho_r(\|x_n - \nabla g^* o T_n(x_n)\|) &\leq D_g(x_n, p) - D_g(v_n, p) \\ &= D_g(x_n, v_n) + D_g(v_n, p) - D_g(v_n, p) + \langle x_n - v_n, \nabla g(v_n) - \nabla g(p) \rangle \\ &\leq D_g(x_n, v_n) + (\|x_n - v_n\|)(\|\nabla g(v_n)\| + \|\nabla g(p)\|) \\ &\leq D_g(x_n, v_n) + M\|x_n - v_n\| \rightarrow 0 \text{ as } n \rightarrow \infty, \end{aligned}$$

where $M = \sup\{\|\nabla g(v_n)\| + \|\nabla g(p)\| : n \in \mathbb{N}\} < \infty$. First, we observe that the sequence $\{\nabla g^* o T_n(x_n)\}_{n \in \mathbb{N}}$ is bounded in E . By Step 2 and the property of ρ_r , in (5.4), we obtain that

$$\lim_{n \rightarrow \infty} \|x_n - \nabla g^* o T_n(x_n)\| = 0.$$

By uniform continuity of ∇g on bounded subsets of E , we get that $\lim_{n \rightarrow \infty} \|\nabla g(x_n) - T_n x_n\| = 0$. Since $\{T_n\}_{n \in \mathbb{N}}$ satisfies the ∇g -NST-condition with Θ , we conclude that $\lim_{n \rightarrow \infty} \|\nabla g(x_n) - Tx_n\| = 0$.

Step 4: $x^* \in F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)$. From Step 3, we have $\lim_{n \rightarrow \infty} \|\nabla g(x_n) - Tx_n\| = 0$ for all $T \in \Theta$. We also know from Step 2 that $x_n \rightarrow x^* \in C$. Since each $T \in \Theta$ is closed, we conclude that $x^* \in F_{\nabla g}(\Theta)$. Furthermore, from Step 2, we obtain $\lim_{n \rightarrow \infty} \|\nabla g(y_n) - \nabla g(v_n)\| = 0$. Since $\{r_n\}_{n \in \mathbb{N}} \subset [a, \infty)$, we obtain that

$\lim_{n \rightarrow \infty} \frac{\|\nabla g(y_n) - \nabla g(v_n)\|}{r_n} = 0$. Considering $v_n = K_{r_n} y_n$ and by Lemma 5.4, we are led to

$$h(\nabla g(v_n), \nabla g(y)) + \frac{1}{r_n} \langle y - v_n, \nabla g(v_n) - \nabla g(y_n) \rangle \geq 0, \quad \forall y \in C.$$

By (C2) of Remark 5.1, we have that

$$\frac{1}{r_n} \langle y - v_n, \nabla g(v_n) - \nabla g(y_n) \rangle \geq h(\nabla g(y), \nabla g(v_n)),$$

Since $\Phi(y) = F(\nabla g(y), \nabla g(y))$ belongs to $\Lambda_1(E)$, we see that $h(\nabla g(y), \nabla g(x^*)) \leq 0, \forall y \in C$. For $t \in (0, 1]$ and $y \in C$, letting $y_t^* = t\nabla g(y) + (1-t)\nabla g(x^*)$, we deduce, by the convexity and closedness of $\nabla g(C)$, that $y_t^* \in \nabla g(C)$. Hence, $h(y_t^*, \nabla g(x^*)) \leq 0, \forall y \in C$. By (C1) in Remark 5.1, we arrive at

$$\begin{aligned} 0 = h(y_t^*, y_t^*) &\leq th(y_t^*, \nabla g(y)) + (1-t)h(y_t^*, \nabla g(x^*)) \\ &\leq h(\nabla g(x^*) + t(\nabla g(y) - \nabla g(x^*)), \nabla g(y)). \end{aligned}$$

Letting $t \downarrow 0$ by (C3), of Remark 5.1, we obtain that $h(\nabla g(x^*), \nabla g(y)) \geq 0$. Thus, $x^* \in \text{BGMEP}(h, \psi, \Phi)$ and by the inclusion $x^* \in F_{\nabla g}(\Theta)$, we arrive at $x^* \in \text{BGMEP}(h, \psi, \Phi)$. This together with $x^* \in F_{\nabla g}(T)$ assures that $x^* \in F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)$.

Step 5: $x_n \rightarrow R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x$. Using Lemma 5.4, we are led to

$$(5.6) \quad D_g(x, R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x) \leq D_g(x, x^*).$$

Also, for $x^* \in F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi) \subseteq C_{n+1}$, $x_{n+1} = R_{C_{n+1}}^g x$, and by Lemma 5.4, we conclude that $D_g(x, x_{n+1}) \leq D_g(x, R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x)$. Since $x_n \rightarrow x^*$ as $n \rightarrow \infty$, we get that $D_g(x, x^*) \leq D_g(x, R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x)$. This together with (5.6) ensures that

$$D_g(x, x^*) = D_g\left(x, R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x\right).$$

By the uniqueness of $R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x$, we obtain $x^* = R_{F_{\nabla g}(\Theta) \cap \text{BGMEP}(h, \psi, \Phi)}^g x$. $\square$

Remark 5.7. Theorem 5.6 is an extension of corresponding one of [21]. In their theorem, the mappings T_n are from a subset $C \subset E$ into the space E , while in Theorem 5.6, the mappings T_n mapping a subset $C \subset E$ into the dual space E^* . Also, Theorem 5.6 addresses the Bregman generalized mixed equilibrium problem (BGMEP), which is not considered in [21]. Finally, the condition $\lim_{n \rightarrow \infty} \beta_n(1 - \beta_n) > 0$, which appears in their result, is not required in Theorem 5.6.

6. CONCLUSION

In this paper, we have provided strong convergence theorems for generalized Bregman nonexpansive mappings in a Banach space. Different from the usual fixed point results, we have also presented a strong convergence theorem for approximating a common ∇g -fixed point of generalized- ∇g -nonexpansive mappings. In future works we will employ our results to optimization, nonlinear variational inequality

and split equilibrium problems in the setting of Banach spaces (see, for some related articles, [8, 17, 25, 26, 35]).

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